

EXECUTIVE SUMMARY

Pesa Salama Classifier

*Real-time Financial Complaint Monitor & Distress Index
for Kenya's Mobile Money Ecosystem*

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50M+

Daily Digital Finance Users

50K

Reviews Analysed

Kenya's Fintech Boom is Outpacing Consumer Protection

Kenya has built one of the world's most advanced digital finance ecosystems yet the system is growing faster than our ability to monitor consumer risk, fraud signals, and service failures in real time.

Consumer Trust Erosion

Unresolved issues erode user confidence in digital finance.

Delayed Crisis Detection

Operational failures identified too late after damage is done.

Regulatory Blind Spots

Authorities lack centralised, real-time ecosystem intelligence.

Revenue & Reputation Risk

Fintech platforms suffer churn and public backlash.

*Millions of complaints posted online annually go unanalysed.
Complaint review is largely manual and reactive.*

How Pesa Salama Works — End to End

01

Data Ingestion

50K+ customer reviews from M-Pesa, KCB, Equity, Tala & Branch (English · Swahili · Sheng)

02

NLP Processing

TF-IDF vectorisation → XGBoost classifier + AfriBERTa transformer for multilingual sentiment

03

Topic Modelling

LDA identifies complaint clusters:
Transaction failures · App crashes · Customer service

04

Distress Scoring

Financial Distress Index (0-100) combining complaint volume, sentiment shift & trend signals

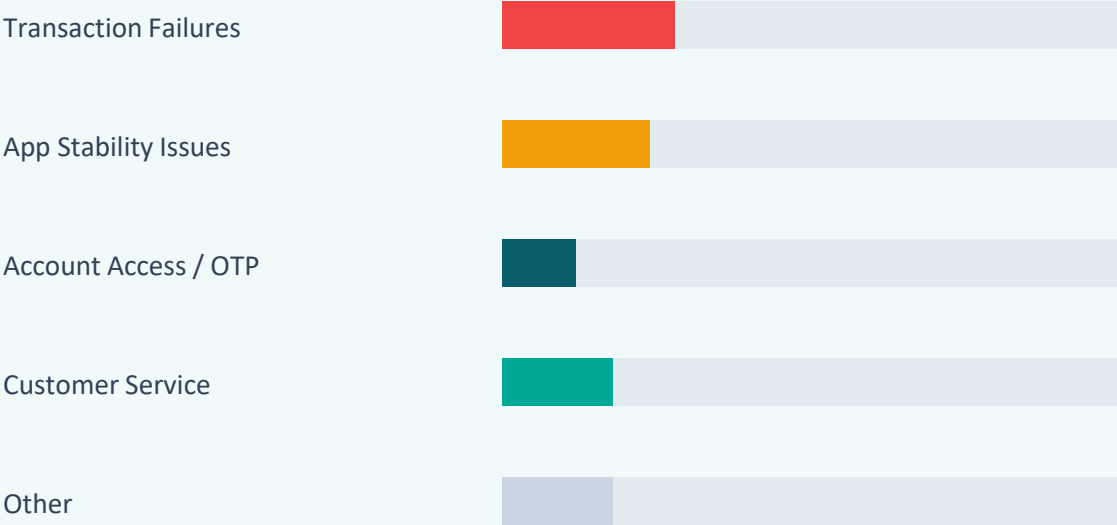
05

Alerting & Dashboards

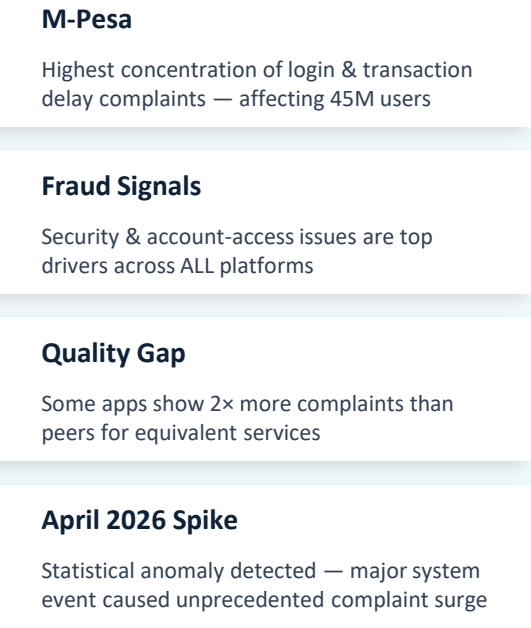
Real-time Streamlit app + Tableau dashboards for regulators & fintech ops teams

What 50,000 Customer Reviews Reveal

Top Complaint Categories



Platform-Level Risks



Machine Learning: Model Evolution & Results

Model	F1-Score	Accuracy	Status
TF-IDF + Logistic Regression (Baseline)	0.827	80%	Baseline
TF-IDF + XGBoost (Intermediate)	0.818	85%	Improved
TF-IDF + XGBoost Tuned (Advanced)	0.848	87%	✓ Production
AfriBERTa Transformer (Multilingual)	0.876	89.5%	✓ Multilingual

Critical Sentiment Signals Identified via SHAP

Negative Drivers: worst · useless · poor · terrible · slow · crashing · login

Positive Drivers: excellent · exceptional · reliable · seamless · efficient

Financial Distress Index (FDI) — Ecosystem Health at a Glance

A single score (0–100) updated daily, combining three real-time signals:

Complaint Pressure

Volume and escalation rate of user complaints

Sentiment Deterioration

Negative shift in customer tone and language

Rising Issue Trends

Increase in recurring problem patterns over time



● Green Zone (0–33)

Low Distress — Stable, customers satisfied

Action: Monitor normally

● Yellow Zone (34–66)

Medium Distress — Issues emerging, complaints rising

Action: Request improvement plan

● Red Zone (67–100)

High Distress — Crisis zone, urgent issues

Action: Emergency intervention

Early Warning Signals & Future Risk Projections

Jan 2023 – Mid 2024	Baseline Stability	Complaint activity predictable and stable. Normal operating baseline.
Late 2024 – 2025	Emerging Friction	Dissatisfaction begins rising steadily. Signals of systemic operational friction.
April 2026	Critical Flashpoint	Severe spike — major system failure or disruptive event detected. Highest z-score recorded.
Post-April 2026	Sustained Elevated Risk	Complaint levels remain above baseline. Issue persists or has lasting effects.

Pesa Salama's forecasting model enables proactive intervention — identifying risks before they escalate into crises.

Four Strategic Actions for Stakeholders

1

Deploy Pesa Salama as an Early-Warning System

Use real-time monitoring and the FDI to identify emerging risks before they escalate into service disruptions, fraud incidents, or reputational crises.

2

Prioritise Service Reliability Improvements

Reduce transaction failures, app crashes, login issues, and OTP delays — together these account for 64% of customer dissatisfaction.

3

Strengthen Fraud Detection & Consumer Protection

Integrate complaint-based fraud signals into operational workflows for faster investigation, response, and customer protection.

4

Establish Data-Driven Regulatory Oversight

Leverage ecosystem-wide dashboards to support proactive supervision, evidence-based interventions, and accountability across all fintech providers.

Deploy Pesa Salama as Kenya's national fintech intelligence platform that enables real-time consumer risk monitoring, earlier intervention, and a safer financial ecosystem.